Shanmukh Upadhyayula

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Education

University of Illinois Urbana-Champaign GPA: 3.95/4.0

Master of Computer Science (MCS), Planned *August 2027 – May 2028*

Bachelor of Science in Computer Science & Geographic Information Science *August 2024 – May 2027*

- Relevant Coursework: Applied Linear Algebra, Data Structures, Algorithms & Models of Computation, Computer Systems, Virtual Reality, Spatial Analysis, Database Management in the Cloud
- Planned: Computational Photography, Computer Vision, Autonomous Vehicle System Engineering

Experience

UIUC CyberGIS Lab Urbana, IL

Machine Learning Engineering Intern *September 2026 – Present*

- Testing and improving I-GUIDE's geospatial AI agent, an LLM system that writes and runs GIS code
- Benchmarking the agent on GeoAnalystBench (48/50 tasks solved, over-planning on 43/48) and identifying flawed expert ground truth that skews LLM scoring
- Uncovering 2 silent failure modes, partial file reads and wrong CRS units causing a 10x area error, by designing trap datasets with known answers

Semantic Segmentation Failure Analysis Urbana, IL

Undergraduate Researcher | shanmukhupad.github.io/portfolio/seg.pdf *May 2026 – August 2026*

- Trained a DeepLabV3-ResNet50 semantic segmentation model in PyTorch on aerial imagery, and wrote an IEEE-format paper presented at a CyberGIS talk
- Diagnosed failure modes through boundary, class, and confidence analysis, finding 2x lower IoU near class edges and high confidence on misclassified pixels
- Cut the error rate from 51% to under 10% on unseen imagery by building label-free uncertainty scoring (entropy, margin, confidence) to filter unreliable pixels
- Built a sliding-window inference tool generating full-resolution uncertainty maps to rank images for review

ACM SIG-CHI | github.com/sigchi-uiuc/digital-diary Urbana, IL

Project Lead *September 2025 – April 2026*

- Built real-time, hands-free mood logging from webcam video via MediaPipe hand landmark tracking and thumb angle
- Added facial-expression mood detection from MediaPipe face blendshapes (smile, frown, brow, cheek)
- Led a 15-person team shipping a Next.js journaling app integrating mood detection, Claude API, and Google Calendar

UIC Electronic Visualization Laboratory | github.com/ShanmukhUpad/curio Chicago, IL

Software Engineering Intern *June 2025 – August 2025*

- Cut processing time by about 30% across hundreds of urban data pipelines by building a Pandas merge tool
- Added dataset merging and multi-tab data views to Curio, a visual data-pipeline tool for researchers

Projects

Street-to-3D: Visual Odometry & 3D Mapping | *C++, OpenCV, Eigen, Python* October 2026 – Present

- Building a monocular visual odometry pipeline in C++ for camera localization, cutting trajectory drift from 33% to 13% on the KITTI benchmark with keyframe selection, 5-point RANSAC pose estimation, and triangulation
- Generalizing to an unseen KITTI sequence at 3.6% drift with no retuning, verified by a trajectory evaluation suite (ATE, RPE, Sim(3) alignment) that matches the open-source evo toolkit
- Calibrating cameras with OpenCV to recover intrinsics within 0.2 px in synthetic tests, the first stage of a phone-video-to-3D-map pipeline

Off the Clock: Standalone VR Game | *C#, Unity XR, Meta Quest 3* April 2026 – May 2026

- Built tracked sword, laser, and mech combat for a Quest 3 game, profiling and fixing in-headset rendering

Technical Skills

Languages: Python, C++, C, Java, R, Typescript, HTML, SQL
Computer Vision & Deep Learning: PyTorch, torchvision, OpenCV, Eigen, MediaPipe, NumPy, ModernGL/GLSL
CV Methods: Segmentation, Visual Odometry, Multi-View Geometry, Camera Calibration, Uncertainty Estimation
Tools: Git, Docker, FastAPI, Linux, Unity, Google Cloud, GeoPandas, ArcGIS Pro